spit out a type declaration d.ts file, which is used to add type declarations to files that don't have them. The emitDecoratorMetadata property is related to experimentalDecorators and governs some decorator-specific output data. The target property specifies what level of JavaScript should be output by the compiler. We have chosen the es2019 level, which should work in all 2022 browsers, but we could choose es6, es2016, es2017, or esNext if we were willing to give up some browser compatibility—esbuild does respect the target property from this file for converting TypeScript syntax.

A number of other options control how strict the compiler is. We've already seen the allowSyntheticDefaultImports option back in Chapter 12, *Using Redux to Manage State*, as necessary for Redux to work properly with TypeScript. In addition, there is a strict option that turns on quite a few stricter checks that are turned off by default. The strict option

- no longer allows the implicit use of any; all types must be declared (this option can be accessed using the property noImplicitAny);
- requires the this variable to be explicitly declared in all internal functions with noImplicitThis;
- includes more strict type checking around metaprogramming functions like call and apply;
- no longer allows null or undefined to be assigned to any value;
- includes stricter type checking of function type matching; and
- requires class properties that can't take null or undefined to be explicitly initialized in the constructor.

Let's see what happens if we turn on some of the strict options in this project. We're going to try and find a balance that allows us to actually compile (our dependence on third-party tools not written in TypeScript may make perfect strictness impossible). And, as I write this, I really don't know what will happen,